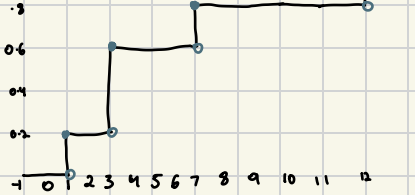


problem set 3

	b	g	gy	r
a	r	0	0	1
	b	1	0	0
	b	1	0	0
	r	0	0	1
	gr	0	1	0
	gy	0	0	1

b. red = $2/6$
blue = $1/3$
green = $1/6$
grey = $1/6$

c. $f(x) = 0 \quad x < 1$
 $f(x) = 0.2 \quad 1 \leq x < 3$
 $f(x) = 0.6 \quad 3 \leq x < 6$
 $f(x) = 0.8 \quad 6 \leq x < 11$
 $f(x) = 1 \quad x \geq 11$



$$2. F_Y(y) = P(a + bY \leq y) = P(X \leq (y-a)/b) = F_X((y-a)/b)$$

$$F_X(u) \cdot \left(\frac{du}{dy}\right) = \left(\frac{y-a}{b}\right) \cdot \left(\frac{1}{b}\right)$$

$$F_Y(y) = \frac{1}{b} \cdot F_X\left(\frac{y-a}{b}\right)$$

$$b. \frac{1+3+4}{3} = \frac{8}{3}$$

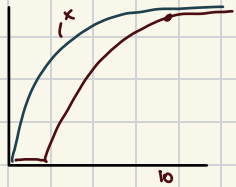
$$c. 1^2 + 3^2 + 4^2 = \frac{1+9+16}{3} = \frac{26}{3} = 8.667$$

$$d. \left(\frac{8}{3}\right)^2 = \frac{64}{9} = 7.111$$

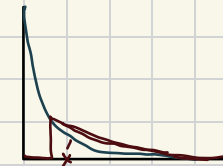
they're not equal

3.

a. Dist. func

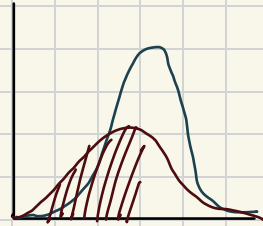
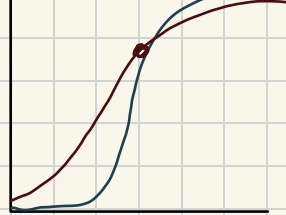


density



4.

Dist func



$$b. x \cdot 3 = e^{-3} = 0.050$$

$$x \leq 3 = 1 - e^{-(3x)} \approx 1$$

$$c. F_Y(y) = F_X\left(\frac{y-1}{3}\right) \quad y \geq 1$$

$$0 < y < 1$$

$$b. x \geq 0.8 = 1 - F(0.8) = 1 - \frac{1}{1+e^{0.8}}$$

$$\frac{1}{1+e^{0.8}} = 0.310$$

$$x \leq 0.3 = \frac{1}{1+e^0} = 0.524$$

$$c. 2x-1 \leq y = P(X \leq \frac{y+1}{2}) =$$

$$F_X\left(\frac{y+1}{2}\right) = \frac{1}{(1+e^{\frac{y+1}{2}})}$$

$$P(Y \leq 0) = F_X\left(\frac{1}{2}\right) = \frac{1}{1+e^{0.5}} = 0.622$$

5.

$$a. F(m) = \int_0^m \lambda e^{(-\lambda t)} dt = (e^{(-\lambda t)})_0^m = 1 - e^{(-\lambda m)}$$

$$1 - e^{(-\lambda m)} = \frac{1}{2} = e^{(-\lambda m)} = \frac{1}{2}$$

$$-\lambda m = \ln\left(\frac{1}{2}\right) = -\ln 2$$

$$m = \frac{\ln 2}{\lambda}$$

$$b. 1 + e^{(-\frac{(m-\mu)}{\sigma})} = 2$$

$$e^{(-\frac{(m-\mu)}{\sigma})} = 1$$

$$-\frac{(m-\mu)}{\sigma} = 0 \quad m = \mu$$

$$d. \lambda \cdot e^{(-\frac{(x-\mu)}{\sigma})} = \frac{1}{4}$$

$$\downarrow = \frac{(1-\mu)}{\sigma}$$

$$-\frac{(x-\mu)}{\sigma} = \ln\left(\frac{(1-\mu)}{\sigma}\right)$$

$$= \ln\left(\frac{1}{4}\right)$$

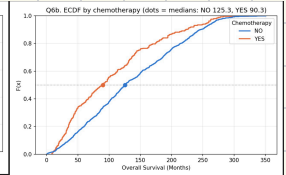
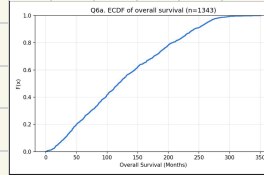
$$F^{-1}(u) = \mu + \sigma \cdot \ln\left(\frac{u}{1-u}\right), 0 < u < 1$$

$$c. e^{(-\lambda x)} = 1 - u$$

$$-\lambda x = \ln(1-u)$$

$$F^{-1}(u) = \frac{-\ln(1-u)}{\lambda} \quad 0 \leq u < 1$$

6.



b. chemo patients have a lower survival rate so we can predict a shorter survival time. the median is 35 months lower, however this prediction also excludes factors like how bad the cancer progression is per person

c. The chemo wasn't randomly assigned so we can't conclude anything from this specific data set. The chemo patients had it none when it came to cancer progression, meaning they were already in bad shape & the chemo was their last hope